

Constrained Molecular Graph Generation with Diffusion Models



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December 2025

1 Motivation & Problem

The Reliability Gap in Generative Models: Why drug discovery requires guaranteed structural constraints?

2 Background

Discrete diffusion on graphs and the ConStruct framework for sampling-time guidance.

3 Proposed Method

Two novel projectors for enforcing ring-based constraints.

4 Experimental Setup

Experimental setup using the QM9 dataset, baseline, and core evaluation metrics.

5 Results

Achieving **100% constraint satisfaction**, analyzing the impact on generative quality, and qualitative examples.

6 Discussion & Conclusion

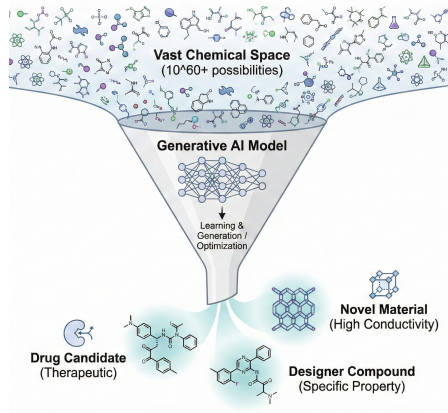
Interpreting the trade-off between strictness and distribution matching, followed by future work.

The Context: Drug Discovery

Designing molecules with specific properties is central to drug discovery and materials science.

- **Drug Discovery:** searching for new therapeutics in an enormous chemical space.
- **Materials Science:** designing compounds with desired physical or chemical behavior.
- **Brute-force fails:** exhaustive search is infeasible; most candidates are invalid.
- **Core Difficulty:** the space of possible molecules is astronomically large and governed by strict structural rules.

Generative AI aims to explore this space efficiently — but structural validity remains a major challenge.



To explore this space with AI, we first need a robust mathematical representation of molecules.

Unlike images, which are continuous grids of pixels, molecules are **discrete structured objects**.

Molecules as Graphs:

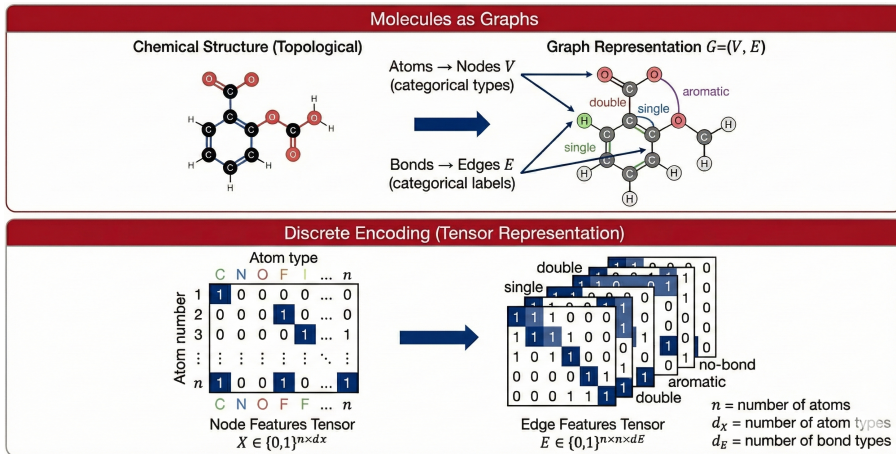
- **Atoms** $\rightarrow$ Nodes V with **categorical** types (e.g., C, N, O, F).
- **Bonds** $\rightarrow$ Edges $E \subseteq V \times V$ with **categorical** labels (e.g., single, double, triple, *no-bond*).

Discrete Encoding (Tensor Representation):

$$X \in \{0, 1\}^{n \times d_X}, \quad E \in \{0, 1\}^{n \times n \times d_E}$$

- n = number of atoms
- d_X = number of atom types
- d_E = number of bond types (including **no-bond**)

The Context: Representation of Molecules as Graphs



Motivation & Problem: The Reliability Gap

What current models do well?

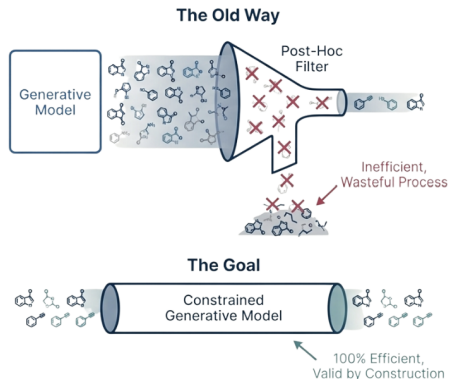
- Current generative models (VAEs, GANs, Diffusion) can learn the **data distribution** and propose diverse candidates.
- They are strong at **pattern learning** (what looks like the dataset), but weak at **rule following** (validity).

The Reliability Gap (Why this is a problem in practice?)

- Generated molecular graphs can be **invalid** (violating structural/chemical constraints).
- Common Workaround: **post-hoc filtering** (e.g., RDKit validity checks) to discard bad/invalid samples.
- **The Issue:** This is inefficient, wasteful (spending compute and time for invalid ones), and provides no feedback to the model.

Goal of this thesis

- Move from *filter after generation* to **constrain during generation**.



Post-hoc filtering wastes samples and does not guarantee constraint satisfaction.

The Challenge: From Soft Guidance to Hard Structural Rules

So the real question is: can we enforce the rules during generation, not after?

Modern generative models learn **statistical correlations**, but many chemical constraints are **non-negotiable** and must hold exactly.

Soft Constraints (Current Practice)

- Conditioning / guidance **nudges** the model toward desired properties.
- Useful for preferences, but **not a guarantee**.
- Violations still happen $\Rightarrow$ you fall back to **post-hoc filtering** (expensive and wasteful).

Hard Constraints (Scientific Need)

- Some rules must hold **exactly** (e.g. valence).
- They must be enforced **inside sampling** so every output is valid **by construction**.

The Need

Shift from *filter after generation* to **constrain during generation**: every generated graph should satisfy the structural rule by construction.

Core Objective

Enable **controllable** and **structurally valid** molecular graph generation by integrating **hard structural constraints** into **discrete diffusion** (focus: ring-based rules).

We model a molecule as an undirected graph $G = (V, E)$.

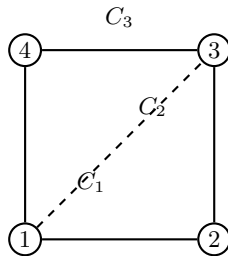
Definition: A **ring** is a **simple cycle** in G : a closed path that visits no vertex more than once (except the start/end vertex).

Why ring structures matter?

- Rings strongly affect stability, reactivity, and drug-likeness.
- Very large rings (macrocycles) are often synthetically challenging.

Key requirements

- **No retraining:** the base diffusion model remains unchanged.
- **No post-hoc rejection:** enforcement happens **during sampling**, not after filtering.
- **Hard guarantee:** every generated graph must satisfy the constraint (**100% compliance**).



*Example: a 4-cycle with a chord contains **three simple cycles** (C_1, C_2, C_3). We count all three as rings.*

Research Questions

- RQ1** How can we enforce strict ring constraints during discrete diffusion process?
- RQ2** How do strict constraints affect validity, diversity, novelty, and distribution alignment?
- RQ3** How does this constrained generation compare to the unconstrained baseline?
-

To answer these, this thesis makes the following key contributions:

- ❶ **Methodology:** Extended the **ConStruct** framework to support complex, structural ring constraints.
- ❷ **Novel Projectors:** Designed sampling-time algorithms for:
 - **Maximum Ring Count** ($\leq K$)
 - **Maximum Ring Length** ($\leq L$)
- ❸ **Validation:** Achieved **100% constraint satisfaction** on QM9 with zero retraining.

Background: Diffusion Models (Standard Intuition)¹

What is “diffusion” (in generative AI)?

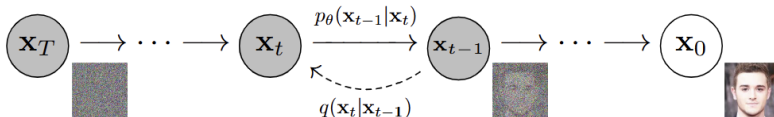
- It is a generation method that learns to **undo noise**.
- First, we define a simple process that **gradually destroys** a real image by adding small random noise many times, until it becomes **almost pure noise**.
- Then we train a neural network to **reverse that process**, i.e., to remove a little noise at each step.
- To generate a new sample, we start from random noise and repeatedly **denoise** until an image appears.

1) Forward process (Diffusion/Corruption) $q(\mathbf{x}_t|\mathbf{x}_{t-1})$

- Add a **small amount of Gaussian noise** each step.
- This part is not learned; it is just a simple corruption.

2) Reverse process (Generation/Denoising) $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$

- Train a neural network to **remove noise step-by-step**.
- Sampling: start from $\mathbf{x}_T$ (noise) and iteratively recover a new $\mathbf{x}_0$ (realistic sample).



Visualizing the forward (dashed) and reverse (solid) processes.

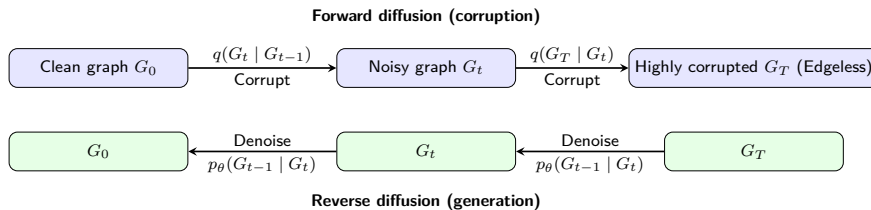
¹J. Ho et al., *Denoising Diffusion Probabilistic Models*. NeurIPS 2020.

Background: The ConStruct Framework² — Discrete Graph Diffusion

Based on a general framework for **discrete denoising diffusion** on graphs.

Why diffusion needs a graph version? $\Rightarrow$ **molecules are discrete**

- Images/audio are **continuous**: you can add a small amount of Gaussian noise.
- Molecular graphs are **categorical**: a bond/atom type is a label, and an edge either exists or not.
- Therefore, it “noises” a graph via **discrete transitions**, by using **categorical Markov transitions**: at each step, nodes/edges may switch categories, and edges may appear/disappear $\Rightarrow$ not Gaussian perturbations.



Forward process: progressively **randomizes** the graph (structure + labels).

Reverse process: a neural network predicts how to **denoise** back toward realistic graphs.

*Goal: generate new graphs that match the training distribution **while enabling constraint enforcement** (next slide)*

²M. Madeira et al., *Generative Modelling of Structurally Constrained Graphs*. NeurIPS 2024.

Core Innovation: Edge-Absorbing (Monotone) Forward Process

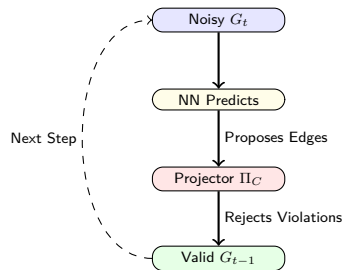
- ConStruct modifies discrete diffusion so the **forward noise is monotone**: each present edge can only transition toward **absorbing “no-edge” state**.
- Therefore, the forward chain **only deletes edges** (never adds new ones).

Sampling-Time Projectors (Hard Guarantees)

- In reverse diffusion, the model proposes a provisional graph $\tilde{G}_{t-1}$ from G_t .
- A projector Π_C **rejects only violating edge additions** and outputs:

$$G_{t-1} = \Pi_C(\tilde{G}_{t-1}, G_t)$$

- This enforces C **at every step** (hard satisfaction), **without retraining**.



Reverse diffusion in ConStruct.

Why constraints can be preserved: Edge-Deletion Invariance

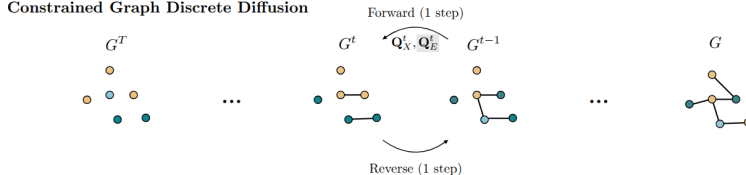
- **Definition:** A constraint C is **edge-deletion invariant** if $G \models C \Rightarrow G' \models C$ for any G' obtained by **deleting edges** from G .

$\Rightarrow$ A constraint is **edge-deletion** invariant if once a graph satisfies the constraint, it will still satisfy it after you remove some edges (bonds). In other words: deleting edges cannot “break” that rule.

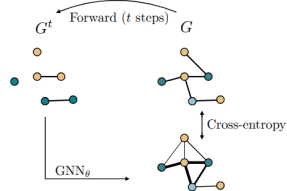
²M. Madeira et al., *Generative Modelling of Structurally Constrained Graphs*. NeurIPS 2024.

The ConStruct Framework: Constrained Graph Discrete Diffusion (Overview)

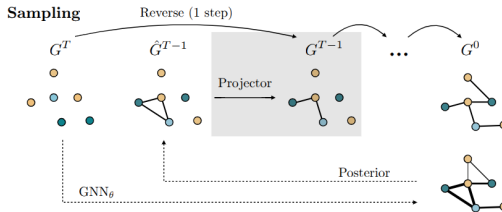
Constrained Graph Discrete Diffusion



Training



Sampling



Schematic overview of ConStruct's constrained discrete diffusion process during training and sampling.

Source: Madeira et al., NeurIPS 2024.

Our Proposed Method: Foundations & Two Novel Ring Projectors

How Our Method Builds on ConStruct

- We adopt ConStruct's **edge-absorbing** discrete diffusion: the forward process **only deletes** bonds.

1. Ring Count $\leq K$

- Constraint: Total number of simple cycles must not exceed K .
- Projector: Rejects any proposed edge that increases the cycle count beyond K .
- Rationale: Controls overall structural complexity.



2. Maximum Ring Length $\leq L$

- Constraint: Every cycle must have length $\leq L$ atoms.
- Projector: Rejects any edge whose addition creates a cycle longer than L .
- Rationale: Avoids chemically implausible or synthetically difficult ring sizes (e.g., avoiding large macrocycles).

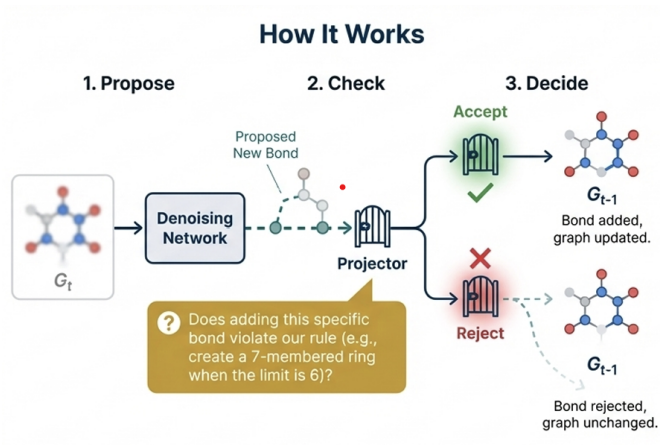


Why It Works:

Both constraints are **edge-deletion invariant**, making them fully compatible with ConStruct's monotone diffusion:

- ✓Deleting edges cannot **create** cycles $\Rightarrow$ ring count cannot increase, and it cannot create a **longer** cycle.
- ✓Therefore, rings can only be introduced by **adding edges during reverse diffusion**.

The Ring Projector: A Smart Gatekeeper Inside the Diffusion Process



The Analogy: Think of it as a gatekeeper that vets every new bond the model proposes to add.

Experimental Setup

Dataset: QM9 $\Rightarrow$ a standard benchmark

- $\sim 130\text{K}$ small organic molecules with up to 9 heavy atoms (C, N, O, and F).
- Represented as **undirected categorical graphs** (our molecular setting).
- Standard train/validation/test split with RDKit sanitization.

Sampling Setup

- **Model:** Single trained ConStruct model (no retraining for constraints).
- **Sampling:** 10,000 molecules per seed $\times$ 5 seeds $\Rightarrow$ **50,000 samples per condition**.
- **Baseline:** Unconstrained sampling (no projector).
- **Constrained runs:** One projector active at a time (ring count or ring length).

Evaluation Metrics

Metric	Description
Validity (%)	RDKit sanitization success rate.
Uniqueness (%)	Fraction of non-duplicate valid molecules.
Novelty (%)	Fraction not found in training set.
FCD	Fréchet ChemNet Distance (distribution match with the dataset).
Disconnected (%)	Fraction of generated graphs with > 1 connected component.
Constraint Sat. (%)	Molecules obeying the rule ($\leq K$ or $\leq L$).

Results: 100% Constraint Satisfaction Across All Settings

Evidence 1: Ring Count ($\leq K$)

K	0-ring	1-ring	2-rings	3-rings	4-rings	5-rings	6-rings	7-rings	8-rings	9-rings	≥ 10
0	100.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	10.996	89.004	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	10.996	72.252	16.752	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	10.996	39.966	10.496	38.542	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	10.996	39.966	7.234	25.012	2.224	0.012	7.904	6.462	0.180	0.010	0.000
Unc.	10.996	39.966	7.234	25.010	2.072	0.012	6.222	5.130	0.138	3.220	0.000
QM9	10.233	39.323	7.369	24.453	2.706	0.026	5.841	6.606	0.329	0.002	3.114

Table: Ring count distributions for constraint ring counts $\leq K$ (%).

Evidence 2: Ring Length ($\leq L$)

L	0-atom	3-atoms	4-atoms	5-atoms	6-atoms	7-atoms	8-atoms	9-atoms	≥ 10
0	100.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	54.342	45.658	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	38.518	29.278	32.204	0.000	0.000	0.000	0.000	0.000	0.000
9	10.996	10.768	10.724	26.590	19.022	11.722	7.956	2.222	0.000
Unc.	10.996	10.768	10.724	26.590	19.022	11.722	7.956	2.222	0.000
QM9	10.233	10.591	11.073	26.934	20.457	11.526	6.921	2.266	0.000

Table: Maximum ring length distributions for constraint ring length $\leq L$ (%).

Results: Impact on Generative Quality (Ring Count Constraint)

How do ring-count constraints affect core generative quality metrics?

K	FCD	Uniqueness (%)	Novelty (%)	Validity (%)	Disconnected (%)
0	10.66 ± 0.02	81.86 ± 0.31	90.68 ± 0.23	99.63 ± 0.03	1.37 ± 0.13
1	1.81 ± 0.01	93.19 ± 0.31	87.29 ± 0.13	99.27 ± 0.14	1.83 ± 0.16
2	1.41 ± 0.01	94.51 ± 0.37	87.22 ± 0.17	99.17 ± 0.14	1.90 ± 0.15
3	1.08 ± 0.01	96.54 ± 0.17	88.55 ± 0.10	98.97 ± 0.11	2.10 ± 0.09
9	1.03 ± 0.01	97.33 ± 0.14	88.74 ± 0.09	98.67 ± 0.11	2.14 ± 0.13
Unc.	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12

Table: Core metrics under ring count constraint $k \leq K$.

Observation: Ring Count ($\leq K$)

- **Validity stays very high:** Validity remains around 99% for all K , peaking at 99.6% for the acyclic case ($K = 0$).
- **FCD jumps:** This measures how close we are to the original dataset distribution. At $K = 0$, FCD jumps to ≈ 10.7 , indicating that enforcing acyclicity moves the model into a ring-free subspace far from the QM9 distribution.
- **Novelty peak:** Novelty is highest for $K = 0$ ($\approx 90.7\%$) and otherwise remains comparable to the baseline.
- **Trade-off in uniqueness:** Uniqueness drops to $\approx 82\%$ at $K = 0$ (much smaller feasible space), but quickly recovers as K increases.

Results: Impact on Generative Quality (Ring Length Constraint)

How do ring-length constraints affect the same core metrics?

L	FCD	Uniqueness (%)	Novelty (%)	Validity (%)	Disconnected (%)
0	10.66 ± 0.02	81.86 ± 0.31	90.68 ± 0.23	99.63 ± 0.03	1.37 ± 0.13
3	5.18 ± 0.03	90.76 ± 0.44	92.35 ± 0.30	99.45 ± 0.06	1.57 ± 0.13
4	3.95 ± 0.01	94.39 ± 0.28	92.95 ± 0.30	98.88 ± 0.06	1.72 ± 0.10
9	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12
Unc.	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12

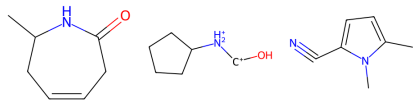
Table: Core metrics under maximum ring length constraint $L_{\max} \leq L$.

Observation: Maximum Ring Length ($\leq L$)

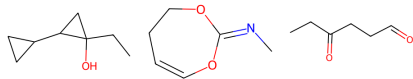
- **Validity remains high:** All configurations preserve high validity ($> 98.4\%$), even under strict ring-size limits.
- **Novelty sweet spot:** Intermediate constraints ($L = 3, 4$) yield the highest novelty ($> 92\%$) $\Rightarrow$ by forbidding the standard 5- and 6-membered rings that dominate the training data, the model is forced to be creative.
- **Controlled distribution shift:** FCD increases smoothly as L decreases, reflecting a gradual move away from the QM9 ring-size distribution as rings are restricted.
- **Recovered uniqueness vs $L = 0$:** Allowing small rings ($L = 3$) already recovers much of the uniqueness ($81.9\% \rightarrow 90.8\%$) lost in the fully acyclic regime $\Rightarrow$ because the space of valid acyclic molecules is smaller.

Qualitative Results: Visualizing Constraints

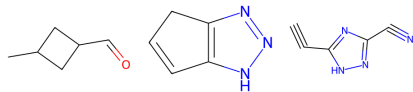
Impact of Ring Count (K)



$K = 1$ (Predominantly monocyclic)

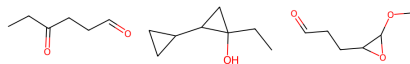


$K = 2$ (Up to two rings per molecule)

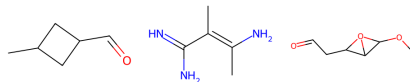


$K = 3$ (Up to three rings per molecule)

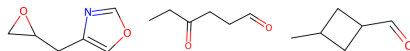
Impact of Ring Length (L)



$L = 3$ (Only very small rings)



$L = 4$ (Up to 4-atom rings)



$L = 5$ (Allows slightly larger cycles)

- **Hard constraints achieved without retraining**

- This makes the method flexible; rules can be changed on the fly without touching the neural network weights.
- The projector enforces the rule **during sampling** (not after), yielding **100% constraint satisfaction**.
- This converts constraints from *soft preferences* into **guarantees by construction**.

- **FCD captures an intentional distribution shift (not “model failure”)**

- The $K = 0$ case is **acyclic** and therefore structurally far from ring-rich QM9.
- The large FCD (≈ 10.66) reflects this **forced move to a different subspace** of chemical space.
- For relaxed settings ($K \geq 1$ or $L = 9$), FCD returns close to the unconstrained baseline (≈ 1.03).

- **Constraints can increase exploration rather than cause collapse**

- Novelty remains high overall and **peaks under moderate ring-length limits** ($L = 3, 4$: $> 92\%$).
- Constraints act as a **structured exploration pressure** $\Rightarrow$ the model explores valid regions it would otherwise under-sample.

- **Side-effect we learn from: fragmentation does not automatically improve**

- Disconnected graphs remain non-trivial (roughly 1.4%–2.1% across settings), even when rings are controlled.
- This indicates disconnection is driven by **broader generation dynamics** (beyond just ring formation), motivating future constraints (e.g., connectivity) or decoder improvements.

Summary & Conclusion

- We extended **ConStruct** with novel projectors for **ring constraints**.
- Demonstrated **hard constraint enforcement during sampling** (not post-hoc filtering, not retraining).
- Showed that discrete diffusion models can be made **strictly constraint-enforcing**.
- Achieved **100% satisfaction** of structural constraints (ring count and ring length) with **no retraining**.

Future Work

- **Hybrid control:** Combining hard structural constraints with property guidance to target both validity and desirability.
- **Pharmacological properties:** Incorporate property constraints (Lipinski's Rule of Five) via sampling-time guidance; investigate hard projection when the property is computable on partial graphs.
- **Joint Constraints:** combining node-level (valency) and graph-level (rings) constraints simultaneously.
- **Scalability:** Evaluating on larger benchmarks (e.g., ZINC/GuacaMol) to stress-test runtime and constraint interaction.

THANK YOU!

Constrained Molecular Graph Generation with Diffusion Models

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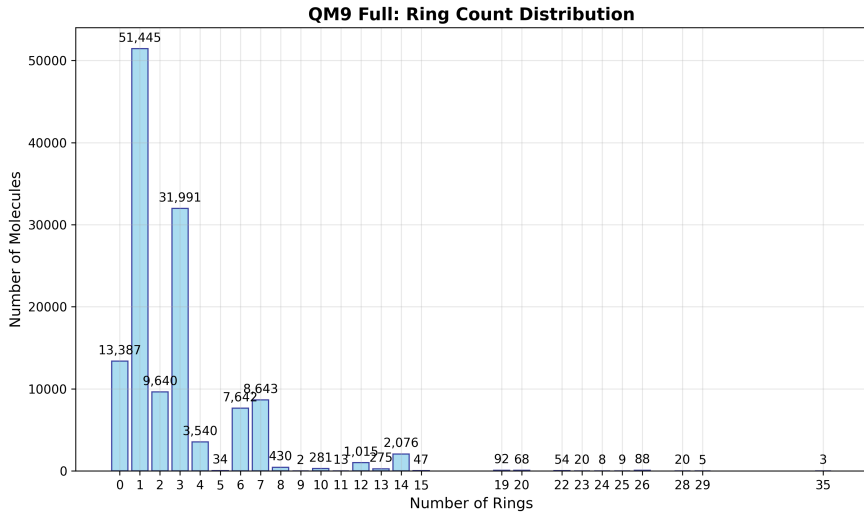
Master Thesis Defense

Code & Resources:

`github.com/ranaislek/ConStruct-Thesis`

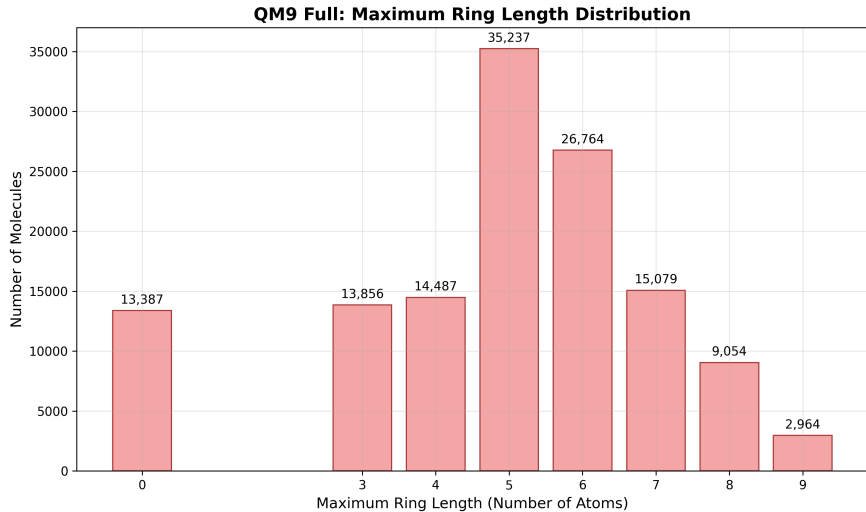
Questions?

Appendix: Dataset Statistics (QM9) — Ring Count



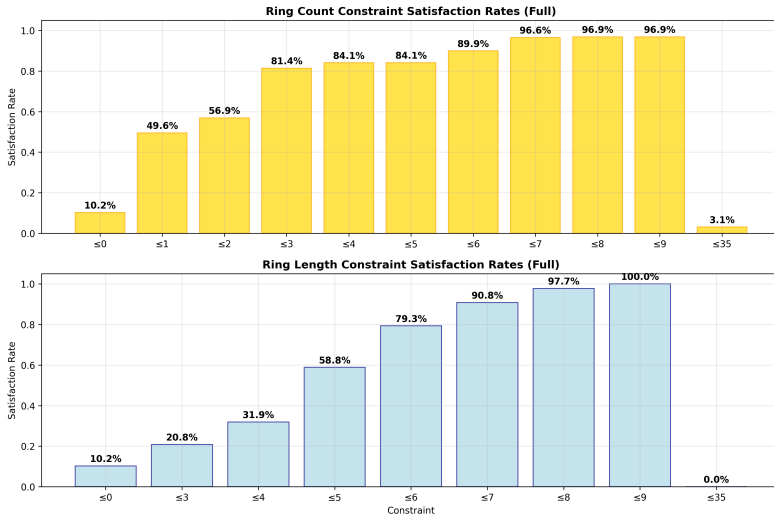
Ring-count distribution in the full QM9 dataset.

Appendix: Dataset Statistics (QM9) — Ring Length



Ring-length distribution in the full QM9 dataset.

Appendix: Constraint Satisfaction Rates (QM9)



Constraint satisfaction rates on the full QM9 dataset for ring-count (top) and ring-length (bottom) constraints.

Ring Count ($\leq K$) - Distribution

K	0-ring	1-ring	2-rings	3-rings	4-rings	5-rings	6-rings	7-rings	8-rings	9-rings	≥ 10
0	100.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	10.996	89.004	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	10.996	72.252	16.752	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	10.996	39.966	10.496	38.542	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	10.996	39.966	8.458	37.196	3.384	0.000	0.000	0.000	0.000	0.000	0.000
5	10.996	39.966	8.458	37.188	3.372	0.020	0.000	0.000	0.000	0.000	0.000
6	10.996	39.966	7.234	29.812	2.714	0.014	9.264	0.000	0.000	0.000	0.000
7	10.996	39.966	7.234	25.012	2.352	0.012	7.908	6.520	0.000	0.000	0.000
8	10.996	39.966	7.234	25.012	2.224	0.012	7.906	6.470	0.180	0.000	0.000
9	10.996	39.966	7.234	25.012	2.224	0.012	7.904	6.462	0.180	0.010	0.000
Unc.	10.996	39.966	7.234	25.010	2.072	0.012	6.222	5.130	0.138	3.220	0.000
QM9	10.233	39.323	7.369	24.453	2.706	0.026	5.841	6.606	0.329	0.002	3.114

Table: Ring count distributions for constraint ring counts $\leq K$ (%).

Ring Count ($\leq K$) - Core Metrics

K	FCD	Uniqueness (%)	Novelty (%)	Validity (%)	Disconnected (%)
0	10.66 ± 0.02	81.86 ± 0.31	90.68 ± 0.23	99.63 ± 0.03	1.37 ± 0.13
1	1.81 ± 0.01	93.19 ± 0.31	87.29 ± 0.13	99.27 ± 0.14	1.83 ± 0.16
2	1.41 ± 0.01	94.51 ± 0.37	87.22 ± 0.17	99.17 ± 0.14	1.90 ± 0.15
3	1.08 ± 0.01	96.54 ± 0.17	88.55 ± 0.10	98.97 ± 0.11	2.10 ± 0.09
4	1.06 ± 0.01	96.73 ± 0.17	88.63 ± 0.08	98.93 ± 0.12	2.11 ± 0.11
5	1.06 ± 0.01	96.73 ± 0.17	88.63 ± 0.08	98.93 ± 0.12	2.11 ± 0.11
6	1.04 ± 0.01	97.11 ± 0.18	88.71 ± 0.08	98.70 ± 0.12	2.13 ± 0.13
7	1.03 ± 0.01	97.33 ± 0.14	88.74 ± 0.09	98.67 ± 0.11	2.14 ± 0.13
8	1.03 ± 0.01	97.33 ± 0.14	88.74 ± 0.09	98.67 ± 0.11	2.14 ± 0.13
9	1.03 ± 0.01	97.33 ± 0.14	88.74 ± 0.09	98.67 ± 0.11	2.14 ± 0.13
40	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12
Unc.	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12

Table: Core metrics under ring count constraint $\leq K$.

Appendix: Maximum Ring Length Distributions Under Constraint ($\leq L$)

Ring Length ($\leq L$) - Distribution

L	0-atom	3-atoms	4-atoms	5-atoms	6-atoms	7-atoms	8-atoms	9-atoms	≥ 10
0	100.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3	54.342	45.658	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	38.518	29.278	32.204	0.000	0.000	0.000	0.000	0.000	0.000
5	17.006	16.818	18.536	47.640	0.000	0.000	0.000	0.000	0.000
6	12.124	12.384	13.354	35.096	27.042	0.000	0.000	0.000	0.000
7	11.264	11.178	11.316	30.436	21.264	14.542	0.000	0.000	0.000
8	11.022	10.824	10.796	27.116	19.474	12.116	8.652	0.000	0.000
9	10.996	10.768	10.724	26.590	19.022	11.722	7.956	2.222	0.000
Unc.	10.996	10.768	10.724	26.590	19.022	11.722	7.956	2.222	0.000
QM9	10.233	10.591	11.073	26.934	20.457	11.526	6.921	2.266	0.000

Table: Maximum ring-length distributions for generated molecules under constraint L .

Ring Length ($\leq L$) - Core Metrics

L	FCD	Uniqueness (%)	Novelty (%)	Validity (%)	Disconnected (%)
0	10.66 ± 0.02	81.86 ± 0.31	90.68 ± 0.23	99.63 ± 0.03	1.37 ± 0.13
3	5.18 ± 0.03	90.76 ± 0.44	92.35 ± 0.30	99.45 ± 0.06	1.57 ± 0.13
4	3.95 ± 0.01	94.39 ± 0.28	92.95 ± 0.30	98.88 ± 0.06	1.72 ± 0.10
5	1.37 ± 0.02	95.80 ± 0.25	88.86 ± 0.23	98.67 ± 0.07	1.98 ± 0.11
6	1.08 ± 0.01	96.77 ± 0.16	88.58 ± 0.09	98.55 ± 0.07	2.13 ± 0.14
7	1.05 ± 0.01	97.23 ± 0.16	88.90 ± 0.06	98.48 ± 0.08	2.15 ± 0.13
8	1.02 ± 0.01	97.39 ± 0.14	88.76 ± 0.09	98.45 ± 0.08	2.14 ± 0.12
9	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12
12	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12
Unc.	1.03 ± 0.01	97.41 ± 0.15	88.72 ± 0.09	98.45 ± 0.09	2.14 ± 0.12

Table: Core metrics under maximum ring length constraint $\leq L$.

Goal: keep *hard guarantees* for feasibility; add *soft bias* for desirability.

- **Hybrid control (Hard + Soft).**

- **Hard:** apply projector after each proposal: $G_{t-1} = \Pi_{\text{struct}}(\tilde{G}_{t-1})$ (valency/rings).
- **Soft:** reweight candidate reverse moves with a score $s(\cdot)$ (QED/SA/Lipinski-penalty)

- **Pharmacological properties (Lipinski).**

- Use as **late-stage guidance** (last k steps) when molecules are near-valid, and properties are meaningful.
- **Hard projection only with safe partial-graph bounds:** e.g., if current molecule weight (MW) already exceeds a max, reject steps that can only increase MW (monotone violation).
- In the Lipinski Rule of Five, a common threshold is $\text{MW} \leq 500 \text{ Da}$ (Daltons), i.e., g/mol.

- **Joint constraints (Valency + Rings).**

- Combine as intersection: $C = C_{\text{valency}} \cap C_{\text{rings}}$.
- **Feasibility-first:** enforce local valency first; then repair/reject ring-violating edge proposals without breaking valency (bounded retries per step).

- **Scalability (ZINC / GuacaMol).**

- Profile overhead: constraint checks per step + wall-clock per sample vs. graph size/ring density.
- Speed-ups: **incremental cycle checks** on new edge, early-exit violations, cache components/shortest-path info.

Appendix: Projector Cycle-Check & Cost

Per reverse step: validate only the *new* edges

- Start from the current accepted graph G_t .
- Extract candidate additions:

$$C = E(\tilde{G}_{t-1}) \setminus E(G_t) \quad (\text{random order})$$

- For each $e = (u, v) \in C$:
 - If u, v are in different components: **accept** (no cycle can be created).
 - Else test by tentative insertion $G' = G \cup \{e\}$:
 - Enumerate **all simple cycles** $\mathcal{C}(G')$ with Johnson's Algorithm.
 - Enforce either constraint:
$$|\mathcal{C}(G')| \leq K \quad \text{and/or} \quad \max_{c \in \mathcal{C}(G')} |c| \leq L$$
 - If violated: **reject** and store e in a **global blocklist**.

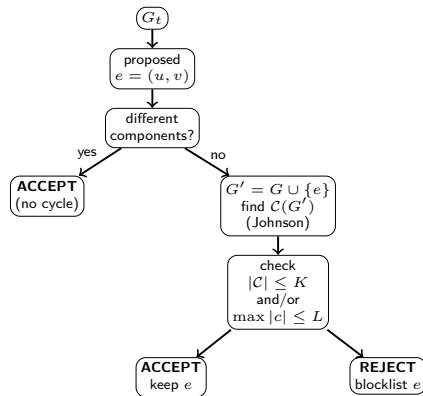
Johnson's Algorithm: DFS/backtracking that lists each *simple* cycle exactly once, using a blocked/unblock set to avoid redundant searches [Johnson, 1975].

Complexity $\Rightarrow$ Worst Case: enumerating all simple cycles is

$$\mathcal{O}((|V| + |E|) \cdot (|\mathcal{C}| + 1))$$

where $|\mathcal{C}|$ is the number of simple cycles in the graph; since $|\mathcal{C}|$ can be exponential in $|V|$, the worst case is exponential.

QM9: $|V| \leq 9$ heavy atoms $\Rightarrow |\mathcal{C}|$ is small in practice $\Rightarrow$ exact cycle checks run in **milliseconds**.



(Schema: per-edge validation inside the projector)

Appendix: Model Definitions (VAE/GAN/Diffusion) & Why Diffusion for Hard Constraints

Goal: enforce *hard, symbolic, non-local* graph constraints **during generation** (not post-hoc).

VAE

- **Definition:** latent-variable generator trained by maximizing an ELBO (reconstruction + KL regularization).
- **Constraint handling:** conditioning / penalties / constrained decoder.
- **Limitation:** constraints are usually *approximate* (no stepwise guarantee) or require filtering.

GAN

- **Definition:** generator trained adversarially against a discriminator (minimax), typically without an explicit likelihood.
- **Constraint handling:** penalties, special architectures, or rejection.
- **Limitation:** training instability / mode collapse; no natural mechanism for guaranteed symbolic constraints.

Diffusion

- **Definition:** iterative generative model that learns to reverse a Markov noising process (denoising steps from noise to data).
- **With ConStruct:** edge-absorbing forward; reverse proposes edge additions.
- **Key advantage:** insert a **projector** each step to **guarantee** edge-deletion-invariant constraints (ring count / ring length) **without retraining**.

Good to know:

- **Why not VAE/GAN?** They can bias toward constraints, but cannot **guarantee** symbolic constraints during generation.
- **Is this rejection sampling?** Not post-hoc: we reject **violating edge additions inside the chain**, so every sampled graph satisfies the structural constraint.

Appendix: DiGress (Discrete Diffusion) — Mechanism & Limitation

Markov assumption (brief): A process is **Markov** if the next state depends only on the current state:

$$p(G_{t+1} \mid G_t, G_{t-1}, \dots) = p(G_{t+1} \mid G_t).$$

In DiGress, this is a **categorical Markov chain** over node/edge types.

Core idea: learn to reverse a discrete noising process on graphs (nodes + edges).

Forward (noise) process:

- Apply categorical Markov transitions to **node types** and **edge types** over timesteps.
- Edges can effectively **appear/disappear** (via transitions), and labels can change.

Reverse (denoise) model (DiGress):

- Uses a **graph transformer** (attention-based GNN) to capture **global** node–node dependencies.
- At each step, predicts **categorical distributions** for **node types** and **edge types** (incl. *no-edge*).

Why hard symbolic constraints are difficult here:

- No **monotonicity** toward a single absorbing structure.
- A constraint can be violated and re-violated across steps; enforcing it typically becomes *approximate guidance* or *post-hoc rejection*.

Appendix: ConStruct — Edge-Absorbing Diffusion & Edge-Deletion-Invariant

ConStruct key design: edge-absorbing forward process (monotone deletion).

Forward (noise):

- Each existing edge transitions toward a **no-edge absorbing state**.
- **Edges are only deleted**; they are never added in the forward chain.

Reverse (denoise / sampling):

- The model proposes an intermediate graph $\tilde{G}_{t-1}$ from G_t (can include new edges).
- Apply a projector: $G_{t-1} = \Pi_C(\tilde{G}_{t-1}, G_t)$
- Π_C **removes only violating proposed edge additions**, so the constraint holds at every step.

Why this yields hard guarantees: **Edge-Deletion Invariance**.

A constraint C is **edge-deletion-invariant** if for any $G \models C$, every G' obtained by **deleting edges** from G also satisfies C .

Ring constraints satisfy this:

- Deleting edges cannot **create** new cycles $\Rightarrow$ ring count cannot increase.
- Deleting edges cannot create a **longer** cycle $\Rightarrow$ max ring length cannot increase.

Implication: Forward diffusion (deletions only) preserves C ; reverse sampling enforces C by rejecting only the edge additions that would violate it $\Rightarrow$ **100% satisfaction**.